Supervised Research
(Law, Economics, and Data Science)
851-0763-00L HS2025

Student: Anna Kravchenko
akravchenko@student.ethz.ch
Student number: 22957856
Outline (10%),
due Nov 21st.

Group member: Anna Kravchenko (solo).
Project Topic: **Anomaly Detection for CNP Transactions.**

Motivation

Online card-not-present payments power modern e-commerce but expose merchants to fast-evolving fraud under strict operational constraints: human review capacity is limited, chargeback labels arrive late, and business teams value stable, actionable alerts over leaderboard metrics. Most academic work optimizes ROC-AUC on randomly shuffled splits, which is a weak proxy for production value because it ignores chronology, alert budgets, and reviewer experience.

In this project will I focus on what actually matters in practice: ranking transactions so that, under a fixed daily review budget, the system consistently surfaces the most useful cases while remaining robust to drift and providing explanations that analysts can trust.

Using the IEEE-CIS / Vesta dataset as a realistic, non-synthetic benchmark, I will build strictly time-ordered training and evaluation windows, compare strong anomaly-detection baselines to a calibrated supervised model, and assess alert quality with budget-aware ranking metrics rather than global discrimination scores.

I am particularly interested in this line of work because it connects rigorous ML with real decision workflows - linking model outputs to reviewer effort, customer friction, and revenue protection - and because it offers a concrete setting to study explainability methods (e.g., SHAP) that can make model decisions transparent for fraud analysts.

Reviewed literature

1. Dal Pozzolo, A., Boracchi, G., Caelen, O., Alippi, C., & Bontempi, G. (2018). *Credit Card Fraud Detection: A Realistic Modeling and a Novel Learning Strategy*. IEEE Transactions on Neural Networks and Learning Systems. URL: https://re.public.polimi.it/bitstream/11311/1044896/1/08038008.pdf
2. Carcillo, F., Le Borgne, Y.-A., Caelen, O., & Bontempi, G. (2018). *Streaming Active Learning Strategies for Real-Life Credit Card Fraud Detection: Assessment and Visualization*. URL: https://arxiv.org/pdf/1804.07481

3. Nguyen, N., Duong, T., Chau, T., Van-Ho Nguyen, Trinh, T., Tran, D., & Ho, T. (2022). *A Proposed Model for Card Fraud Detection Based on CatBoost and Deep Neural Network*. IEEE Access. URL: https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9882043
4. Tian, Y., Liu, G., Wang, J., & Zhou, M. (2023). *Transaction Fraud Detection via an Adaptive Graph Neural Network.* URL: https://arxiv.org/pdf/2307.05633
5. Li, X., Peng, Y., Sun, X., Duan, Y., Fang, Z., & Tang, T. (2025). *Unsupervised Detection of Fraudulent Transactions in E-commerce Using Contrastive Learning.* URL: https://arxiv.org/pdf/2503.18841
6. Manoharan, G., Ali, S. D. N. H., Sathe, M., Karthik, A., Nagpal, A., & Sidana, A. (2024). *Fraud Detection in E-commerce Transactions: A Machine Learning Perspective.* In *Proceedings of the 2024 International Conference on Electrical, Computer and Energy Systems* (IEEE). URL: https://ieeexplore.ieee.org/document/10601813
7. Molnar, C. (2022). *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable* (2nd ed.). URL: https://christophm.github.io/interpretable-ml-book/

Data

Source: IEEE-CIS / Vesta Fraud Detection - real (non-synthetic) CNP transactions provided by Vesta Corporation for an official IEEE Computational Intelligence Society challenge (hosted on Kaggle).

- `train_transaction.csv` - transaction features + binary target `isFraud`.
- `train_identity.csv` - device/browser/network identity features.
- Join key: `TransactionID` - identity exists for a subset only.
- Time: `TransactionDT` (time delta) used to reconstruct event order.

The work uses the labeled training files `train_transaction.csv` and `train_identity.csv`, with identity coverage ≈24% of transactions.

The unlabeled competition `test` files are excluded from all analyses, since they contain no ground-truth labels and therefore cannot be used for evaluation or for constructing chronological validation splits.

Approach

The study focuses on fraud/anomaly detection for card-not-present (CNP) transactions under practical constraints and will evaluate models through a top-K alerting lens. I will work only with the labeled training tables `train_transaction.csv` (target `isFraud`) and `train_identity.csv`, joined by `TransactionID`. Unlabeled competition test files are excluded. `TransactionDT` is treated as event order (timedelta) to reconstruct a timeline and create strictly chronological train/validation/test splits.

On preparation stage , I will left-join identity to transactions on `TransactionID`, keep at most one record per `TransactionID`, and add explicit missing-value indicators before any imputation.

Basic EDA will cover class imbalance, the fraud rate over time (weekly/monthly view), key feature distributions, and a simple missingness heatmap to understand coverage.

Preprocessing will be fit on the training window only and then applied to validation/test to avoid temporal leakage. Numericals will be imputed and standardized; categoricals will be imputed and encoded with a leakage-safe scheme. The steps will be packaged as a reproducible scikit-learn pipeline.

Modeling will prioritize unsupervised detectors. I will implement Isolation Forest on the preprocessed features and a small tabular autoencoder that uses reconstruction error as the anomaly score. For context, I will also train one simple supervised reference model and clearly label it as a baseline; its purpose is only to contextualize unsupervised scores, not to chase leaderboard metrics.

Evaluation will follow the operational scenario: for each day in the validation/test windows, score and rank transactions and report top-K alert quality using Precision@K and Recall@K (with K set to a few realistic daily budgets). I will also track the stability of these metrics across time and keep ROC-AUC as a secondary, descriptive number.